

# Supplementary Material

## 1 PARTICIPANTS INFORMATION

**Table S1.** Clinical characteristics and MDS-UPDRS Part III scores obtained during the recording sessions across 18 participants.

| Measure                                                  | OFF medication                | ON medication            | OFF-ON change        |
|----------------------------------------------------------|-------------------------------|--------------------------|----------------------|
| MDS-UPDRS Part, mean $\pm$ SD (range)                    | 37.3 $\pm$ 17.3 (10–75)       | 28.0 $\pm$ 15.0 (7–61)   | 9.3 $\pm$ 4.2 (3–17) |
| Hoehn & Yahr stage, median (range)                       | 2 (1–4)                       | 2 (0–3)                  | –                    |
| Time since last dopaminergic dose, mean $\pm$ SD (range) | 1033.6 $\pm$ 514.3 (720–3006) | 68.2 $\pm$ 16.0 (36–102) | –                    |
| Dyskinesias present during rating,                       | 1/18                          | 3/18                     | –                    |

**Table S2.** Behavioral descriptive summary by medication and run. Raw trials were 90 per subject per run. The number of subjects was X. Retained trials are reported as median [range] across subjects; RT measures are reported as mean  $\pm$  SD across subjects.

| Medication | Run | Retained trials | Mean RT (ms) | RT CV     | Consecutive RT variability |
|------------|-----|-----------------|--------------|-----------|----------------------------|
| OFF        | 1   | X [X, X]        | X $\pm$ X    | X $\pm$ X | X $\pm$ X                  |
| OFF        | 2   | X [X, X]        | X $\pm$ X    | X $\pm$ X | X $\pm$ X                  |
| ON         | 1   | X [X, X]        | X $\pm$ X    | X $\pm$ X | X $\pm$ X                  |
| ON         | 2   | X [X, X]        | X $\pm$ X    | X $\pm$ X | X $\pm$ X                  |

**Table S3.** Fixed-effect estimates from the linear mixed-effects model predicting reaction time.

| Predictor    | Contrast / unit    | Estimate $\beta$ | 95% CI | p-value |
|--------------|--------------------|------------------|--------|---------|
| Medication   | ON – OFF           | X ms             | [X, X] | X       |
| GVS pooled   | Active – Sham      | X ms             | [X, X] | X       |
| Run          | Run 2 – Run 1      | X ms             | [X, X] | X       |
| Trial number | Per trial increase | X ms             | [X, X] | X       |

### 1.1 Reward Effects on Reaction Time

We examined whether the reward condition was associated with reaction time (RT). High-reward trials were those assigned a reward value of 5. Low-reward trials were the lower non-negative reward condition, coded as 0 for 5 participants and 1 for 13 participants. Because this low-reward value varied across

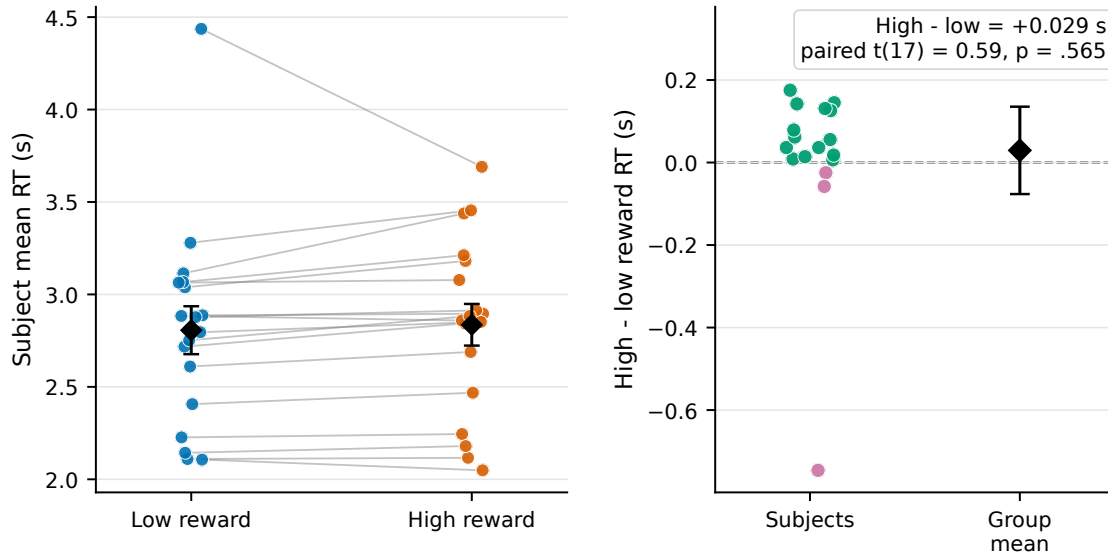


Figure S1: Reward effects on raw RT. Paired participant-level mean RTs are shown for low- and high-reward trials, with black diamonds indicating the group mean  $\pm$  SEM. The high-minus-low reward differences are shown alongside the group mean difference and 95% confidence interval. The raw RT contrast did not show a significant reward effect,  $M_{\text{diff}} = +0.029$  s,  $t(17) = 0.59$ ,  $p = .565$ .

participants, reward was analyzed as a within-participant categorical contrast between low- and high-reward trials. For each participant, mean RT was estimated separately for the two reward conditions, and the high-minus-low difference was tested at the group level with a two-sided paired-samples  $t$ -test.

High-reward trials were not associated with a reliable change in raw RT. Participants were numerically slower on high-reward than on low-reward trials ( $M_{\text{diff}} = +0.029$  s), but this difference was not statistically significant,  $t(17) = 0.59$ ,  $p = .565$ .

## 2 VALIDATION OF THE OPTIMIZATION FRAMEWORK

### 2.1 Model-ablation analysis

Figure S2 compares ablation models using a balanced evaluation criterion that rewards high held-out correlation with behaviour and penalizes instability between train and test. The task term favours components with stronger task-related beta magnitude, the bold and beta terms penalize trial-to-trial variability in BOLD and beta representations, and the smooth term penalizes spatially rough weight maps.

The best overall model has four terms, with the lowest mean evaluation score ( $S = 0.0164$ ). The evaluation score is:

$$S = -w_1 \left\| \text{corr}_{\text{test}} \right\| + w_2 \left\| \left| \text{corr}_{\text{train}} \right| - \left| \text{corr}_{\text{test}} \right| \right\| + w_3 \left\| \frac{\left| \text{loss}_{\text{test}} - \text{loss}_{\text{train}} \right|}{\left| \text{loss}_{\text{train}} \right| + \varepsilon} \right\|$$

The weights  $w_1$ ,  $w_2$  and  $w_3$  are chosen as the inverses of the mean ranges of the corresponding terms, so that each term contributes approximately equally to the final score. Importantly, it is not the model with the highest mean test correlation; the task-only model (task = 0.8, bold = 0, beta = 0, smooth = 0) had the highest test correlation (0.1918) but a much larger relative train–test loss gap (0.2844 vs. 0.1257). The winning model is therefore best interpreted as the most balanced and most stable setting, not simply the strongest correlation setting. Statistically, it was significantly better than 8 of the 11 alternatives, but it was not significantly different from the top three single-term baselines after Bonferroni correction. The combined setting therefore gives the best overall trade-off between behavioural relevance and cross-fold stability, while the advantage over the strongest single-penalty baselines is modest.



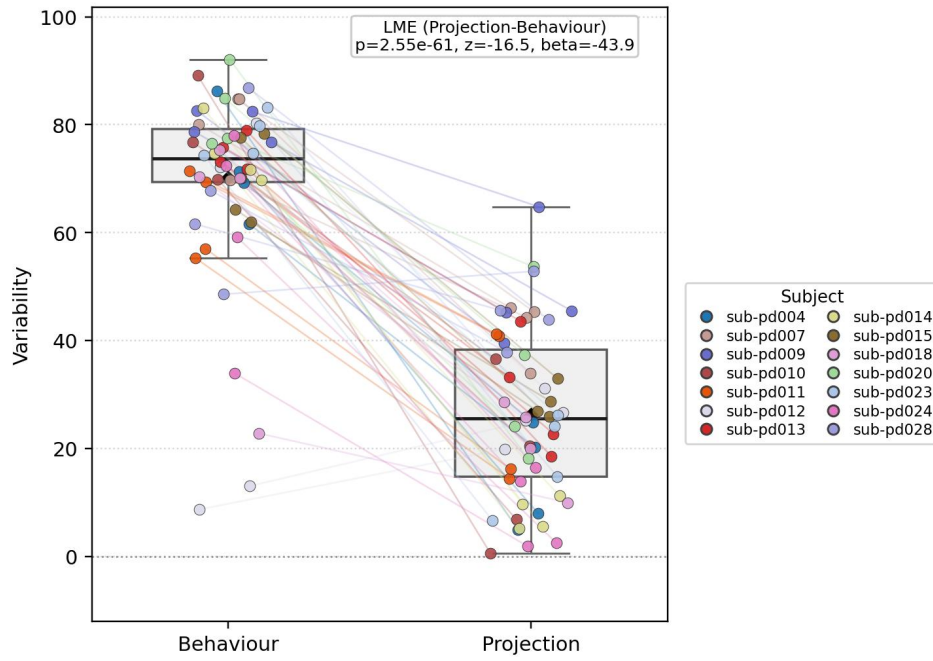


Figure S3: Comparison of trial-to-trial variability in reaction time and the projection signal. Each colored pair represents one run, with colors indicating subjects. Across 56 runs from 14 subjects, the optimized network score showed lower variability than reaction time (linear mixed-effects model with subject as a random intercept:  $\beta = -43.94$ ,  $z = -16.52$ ,  $p = 2.55 \times 10^{-61}$ ).

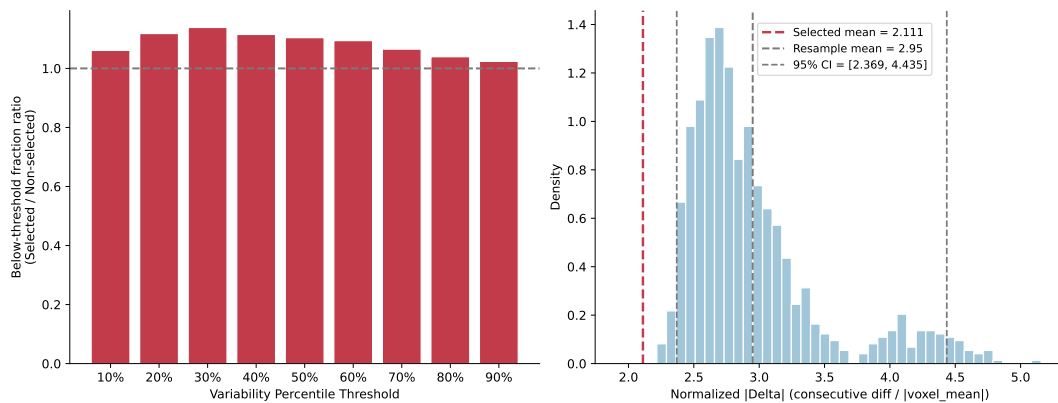


Figure S4: Trial-to-trial variability in the selected voxel network relative to motor-area non-selected voxels. Left, the ratio of voxels falling below pooled percentile thresholds of normalized consecutive-trial difference (selected / non-selected), where values above 1 indicate enrichment of selected voxels in the low-variability tail. Right, the null distribution of variability from 1,000 size-matched resamples of motor-area non-selected voxels, with the observed selected-network mean marked for comparison.

### 3 ANATOMICAL CHARACTERIZATION OF THE FINAL VIGOUR-RELATED NETWORK

#### 3.1 ROI-wise composition of the final network

Anatomical labels correspond to the three Harvard–Oxford labels.

Table S4: ROI-wise composition of the final vigour-related network, sorted alphabetically by ROI name.

| ROI                           | ROI size | Selected voxels | % ROI selected | % of map |
|-------------------------------|----------|-----------------|----------------|----------|
| Left Amygdala                 | 306      | 147             | 48.04          | 5.50     |
| Right Amygdala                | 366      | 74              | 20.22          | 2.77     |
| Right Crus II                 | 1765     | 108             | 6.12           | 4.04     |
| Left Frontal Medial Cortex    | 424      | 21              | 4.95           | 0.79     |
| Right Frontal Medial Cortex   | 444      | 46              | 10.36          | 1.72     |
| Left Frontal Pole             | 7061     | 237             | 3.36           | 8.86     |
| Right Frontal Pole            | 8196     | 194             | 2.37           | 7.26     |
| Left Hippocampus              | 691      | 127             | 18.38          | 4.75     |
| Right Hippocampus             | 700      | 77              | 11.00          | 2.88     |
| Left Inferior Temporal Gyrus  | 1691     | 227             | 13.42          | 8.49     |
| Left Insular Cortex           | 1160     | 2               | 0.17           | 0.07     |
| Left Lateral Occipital Cortex | 4955     | 46              | 0.93           | 1.72     |
| Left Middle Temporal Gyrus    | 2123     | 398             | 18.75          | 14.88    |
| Left Parahippocampal Gyrus    | 978      | 125             | 12.78          | 4.67     |
| Right Parahippocampal Gyrus   | 980      | 56              | 5.71           | 2.09     |
| Left Postcentral Gyrus        | 3628     | 79              | 2.18           | 2.95     |
| Left Precentral Gyrus         | 4296     | 42              | 0.98           | 1.57     |
| Left Precuneus Cortex         | 2478     | 53              | 2.14           | 1.98     |
| Left Putamen                  | 799      | 116             | 14.52          | 4.34     |
| Left Superior Parietal Lobule | 1449     | 129             | 8.90           | 4.82     |
| Left Superior Temporal Gyrus  | 877      | 11              | 1.25           | 0.41     |
| Left Temporal Fusiform Cortex | 873      | 96              | 11.00          | 3.59     |
| Left Temporal Pole            | 2350     | 84              | 3.57           | 3.14     |
| Right VIIb                    | 878      | 43              | 4.90           | 1.61     |
| Right VIIa                    | 853      | 10              | 1.17           | 0.37     |

After normalizing by ROI size, the strongest selection is concentrated in temporal–limbic regions and the left putamen, while motor-related regions such as the superior parietal lobule, cerebellum, and peri-central cortex are also represented but with lower ROI-normalized coverage. In other words, the network has a motor-related core, but its strongest ROI-level enrichment is not confined to the primary motor cortex. It is a distributed, left-biased vigour-related system rather than a purely focal motor map.

### 3.2 Comparison with the task-engagement reference map

The tables below provide a compact summary of the limited overlap between the final vigour-related network and the task-engagement reference map, including both global overlap metrics and region-wise voxel counts.

Table S5. Comparison of the full-constraint network and the task-engagement reference map.

| Network                 | Voxel count | Shared voxels (% of network) | Main anatomical labels                                                                               |
|-------------------------|-------------|------------------------------|------------------------------------------------------------------------------------------------------|
| Reference map           | 2674        | 102 (3.81%)                  | Frontal Pole; Middle Temporal Gyrus, posterior division; Inferior Temporal Gyrus, posterior division |
| Postcentral-boosted map | 1505        | 102 (6.78%)                  | Lateral Occipital Cortex, superior division; Occipital Pole; Cingulate Gyrus, posterior division     |

Pairwise metrics: Dice coefficient = 0.049; center-of-mass distance = 47.3 mm.

Table S6: Regional voxel counts of the full-constraint and active networks.

| Region                                                                  | Full-constraint network | Task-activation network |
|-------------------------------------------------------------------------|-------------------------|-------------------------|
| <i>Regions with greater contribution in the full-constraint network</i> |                         |                         |
| Frontal Pole                                                            | 431                     | 0                       |
| Middle Temporal Gyrus                                                   | 398                     | 24                      |
| Inferior Temporal Gyrus                                                 | 227                     | 0                       |
| Parahippocampal Gyrus                                                   | 181                     | 118                     |
| Left Amygdala                                                           | 135                     | 0                       |
| Superior Parietal Lobule                                                | 129                     | 0                       |
| Left Hippocampus                                                        | 123                     | 43                      |
| Left Putamen                                                            | 116                     | 0                       |
| Left Cerebral White Matter                                              | 115                     | 55                      |
| Temporal Fusiform Cortex                                                | 96                      | 59                      |
| Temporal Pole                                                           | 84                      | 0                       |
| Right Hippocampus                                                       | 69                      | 39                      |
| Frontal Medial Cortex                                                   | 67                      | 0                       |
| Right Amygdala                                                          | 60                      | 0                       |
| Precuneous Cortex                                                       | 53                      | 0                       |
| Left Cerebral Cortex                                                    | 44                      | 10                      |
| Superior Temporal Gyrus                                                 | 11                      | 0                       |
| Insular Cortex                                                          | 2                       | 0                       |
| <i>Regions with greater contribution in the task-activation network</i> |                         |                         |
| Lateral Occipital Cortex                                                | 46                      | 207                     |
| Unlabeled / outside atlas                                               | 161                     | 194                     |
| Occipital Pole                                                          | 0                       | 148                     |
| Cingulate Gyrus                                                         | 0                       | 140                     |
| Right Cerebral White Matter                                             | 5                       | 127                     |
| Postcentral Gyrus                                                       | 79                      | 84                      |
| Central Opercular Cortex                                                | 0                       | 77                      |
| Precentral Gyrus                                                        | 42                      | 55                      |
| Right Thalamus                                                          | 0                       | 50                      |
| Angular Gyrus                                                           | 0                       | 34                      |
| Parietal Opercular Cortex                                               | 0                       | 19                      |
| Right Cerebral Cortex                                                   | 0                       | 11                      |
| Supramarginal Gyrus                                                     | 0                       | 7                       |
| Cuneal Cortex                                                           | 0                       | 2                       |
| Left Thalamus                                                           | 0                       | 2                       |

## 4 MEDICATION-STATE ANALYSES

### 4.1 Subject-wise between-session changes in ROI-edge connectivity

This heatmap shows how the strongest connectivity changes vary across subjects for the most changed ROI edges. Warm colors indicate positive values, meaning higher mutual information in the ON-medication session than in the OFF-medication session, whereas cool colors indicate negative values, meaning lower mutual information in the ON-medication session. The selected edges were ranked by mean absolute between-session change across the 14 paired subjects, so the figure emphasizes edges with the largest subject-level modulation.

### 4.2 Control analysis: no evidence for preferential ON-state stabilization

To test whether the higher ON–ON similarity could be explained by the optimization favouring voxel populations that were intrinsically more stable in the ON state, we examined subject-level normalized absolute consecutive-trial beta differences ( $|\Delta|/|\text{mean}|$ ), where lower values indicate greater stability. In

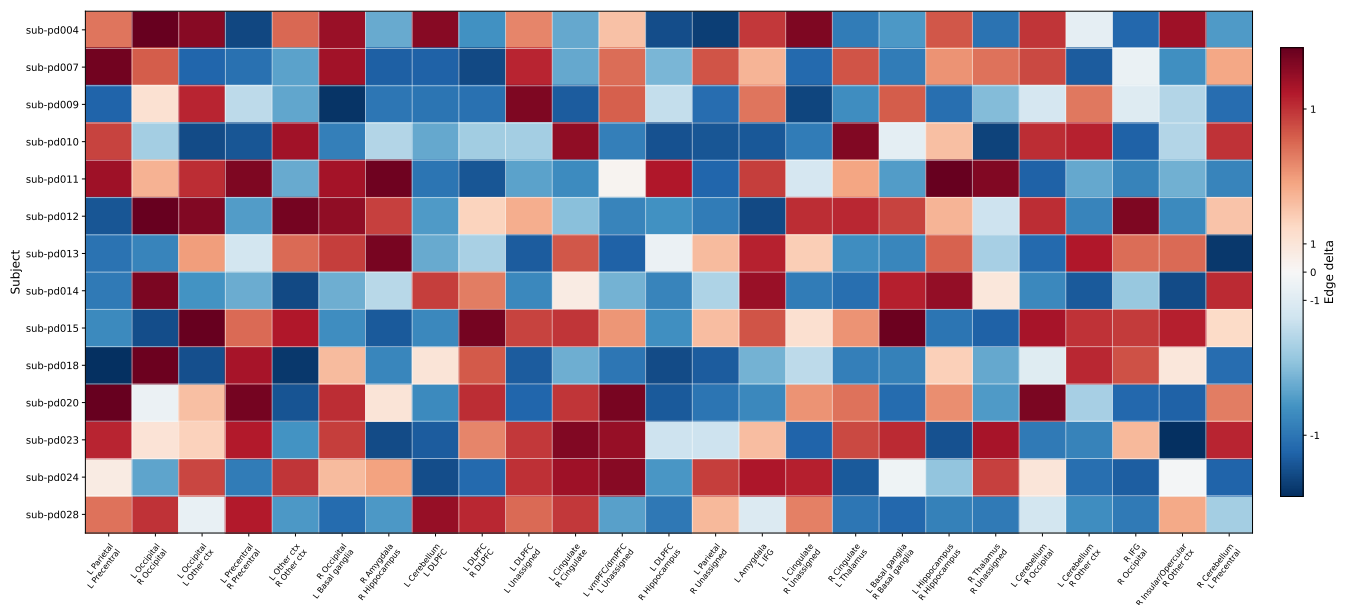


Figure S5: Subject-wise heatmap of the top between-session changes in region-of-interest (ROI) edge connectivity, measured with continuous mutual information estimated using the Kraskov-Stogbauer-Grassberger (KSG) nearest-neighbor estimator. Each row represents one participant, and each column represents one of the top ROI-to-ROI edges selected based on the largest mean absolute change between sessions across all subjects. Values show session 2 minus session 1. Abbreviations: L, left; R, right; DLPFC, dorsolateral prefrontal cortex; vmPFC, ventromedial prefrontal cortex; dmPFC, dorsomedial prefrontal cortex; IFG, inferior frontal gyrus; ctx, cortex.

the paired sample ( $n = 14$ ; Figure S6), the selected voxel set showed lower variability than the motor control pool in both OFF (2.00 vs. 2.52) and ON (2.31 vs. 3.52). Crucially, the selected set did not show preferential stabilization in the ON state; instead, its variability increased only modestly from OFF to ON ( $\Delta_{\text{sel}} = +0.308$ , paired  $t = 0.669$ ,  $p = 0.515$ ). The control pool showed a larger increase ( $\Delta_{\text{ctl}} = +0.998$ , paired  $t = 1.813$ ,  $p = 0.093$ ), and the difference in OFF-to-ON change between the two sets was significant ( $\Delta_{\text{sel}} - \Delta_{\text{ctl}} = -0.690$ ,  $p = 0.046$ ). These results suggest that the optimization procedure favours generally stable voxels, but not voxels that are selectively stabilized in the ON state. Therefore, the stronger ON-ON similarity is unlikely to be explained mainly by an ON-state stability bias and is more consistent with dopaminergic reorganization across patients.

## 5 SUPPLEMENTARY GVS-RELATED ROI ANALYSES

### 5.1 Significant ROI changes by subject and stimulation waveform

This section reports the subject-level sham-referenced ROI effects of active GVS within the final network.

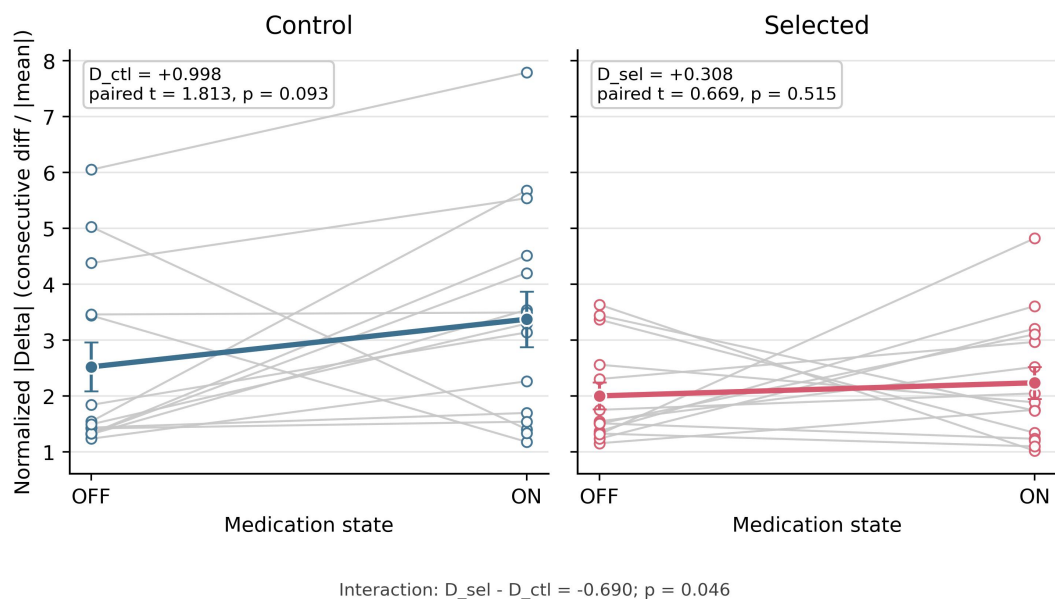


Figure S6: Subject-level OFF and ON values of normalized absolute consecutive-trial beta difference ( $|\Delta|/|\text{mean}|$ ) for the control and selected voxel sets. Lower values indicate greater stability. Gray lines show individual subjects; colored markers and error bars show mean  $\pm$  SEM. The selected set remained less variable than the control set in both states, but did not show preferential stabilization in ON. Instead, the OFF-to-ON increase in variability was smaller in the selected set ( $\Delta_{sel} = +0.308$ ) than in the control set ( $\Delta_{ctl} = +0.998$ ), yielding a significant interaction ( $\Delta_{sel} - \Delta_{ctl} = -0.690, p = 0.046$ ).



Table S7: Significant ROIs by subject and GVS session in the OFF-sham OFF comparison ( $FDR < 0.05$ ), excluding Unassigned Active Voxels. **Purple** text indicates ROIs also significant in the ON-sham ON comparison. DLPC = dorsolateral prefrontal cortex.

| Subject          | GVS1                                                                                                                             | GVS2                                                                  | GVS3                                                                                                                                      | GVS4                                                                    | GVS5                                                                                                                                                 | GVS6                                                                                                                           | GVS7                                                                                              | GVS8                                                                                                                                                       |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>sub-pd004</b> | Cingulate Cortex<br>Precentral                                                                                                   | Precentral<br>Occipital Cortex                                        | -                                                                                                                                         | Precentral                                                              | Cingulate Cortex<br>DLPC                                                                                                                             | Basal Ganglia                                                                                                                  | DLPC<br>Parietal Cortex<br>Precentral<br>Basal Ganglia                                            | Amygdala<br>Occipital Cortex                                                                                                                               |
| <b>sub-pd007</b> | -                                                                                                                                | Inferior Frontal Gyrus<br>DLPC<br>Temporal Cortex<br>Occipital Cortex | DLPC                                                                                                                                      | -                                                                       | Cerebellum<br>DLPC                                                                                                                                   | Amygdala<br>vmPFC/dmPFC                                                                                                        | Cerebellum<br>Parietal Cortex<br>Insular/Opercular Cortex                                         | Cerebellum<br>Inferior Frontal Gyrus<br>Insular/Opercular Cortex                                                                                           |
| <b>sub-pd009</b> | Cerebellum<br>Inferior Frontal Gyrus<br>DLPC                                                                                     | Cerebellum<br>Other Cerebral Cortex                                   | Cerebellum<br>Cingulate Cortex<br>DLPC<br>Parietal Cortex<br>Precentral<br>Temporal Cortex<br>Basal Ganglia                               | Cerebellum<br>DLPC<br>vmPFC/dmPFC<br>Precentral<br>Basal Ganglia        | Cerebellum<br>Cingulate Cortex<br>DLPC<br>vmPFC/dmPFC<br>Temporal Cortex<br>Basal Ganglia<br>Other Cerebral Cortex                                   | Parietal Cortex<br>Temporal Cortex<br>Other Cerebral Cortex                                                                    | Cerebellum<br>DLPC<br>Temporal Cortex<br>Basal Ganglia<br>Other Cerebral Cortex                   | Cerebellum<br>Cingulate Cortex<br>Temporal Cortex<br>Basal Ganglia                                                                                         |
| <b>sub-pd010</b> | -                                                                                                                                | -                                                                     | -                                                                                                                                         | -                                                                       | -                                                                                                                                                    | -                                                                                                                              | -                                                                                                 | -                                                                                                                                                          |
| <b>sub-pd011</b> | -                                                                                                                                | -                                                                     | -                                                                                                                                         | -                                                                       | -                                                                                                                                                    | -                                                                                                                              | -                                                                                                 | -                                                                                                                                                          |
| <b>sub-pd012</b> | -                                                                                                                                | -                                                                     | -                                                                                                                                         | -                                                                       | -                                                                                                                                                    | -                                                                                                                              | -                                                                                                 | -                                                                                                                                                          |
| <b>sub-pd013</b> | Cingulate Cortex<br>Thalamus<br>Occipital Cortex<br>Insular/Opercular Cortex                                                     | vmPFC/dmPFC<br>Thalamus<br>Occipital Cortex                           | Precentral<br>Insular/Opercular Cortex<br>Basal Ganglia                                                                                   | Insular/Opercular Cortex                                                | -                                                                                                                                                    | -                                                                                                                              | Cingulate Cortex                                                                                  | Insular/Opercular Cortex                                                                                                                                   |
| <b>sub-pd014</b> | Precentral<br>Occipital Cortex                                                                                                   | vmPFC/dmPFC<br>Occipital Cortex<br>Other Cerebral Cortex              | Cerebellum<br>Inferior Frontal Gyrus<br>Precentral<br>Thalamus<br>Occipital Cortex<br>Insular/Opercular Cortex<br>Other Cerebral Cortex   | Cerebellum<br>DLPC<br>Insular/Opercular Cortex<br>Other Cerebral Cortex | Cerebellum<br>vmPFC/dmPFC<br>Precentral<br>Temporal Cortex<br>Occipital Cortex<br>Insular/Opercular Cortex<br>Basal Ganglia<br>Other Cerebral Cortex | Inferior Frontal Gyrus<br>Precentral<br>Occipital Cortex<br>Insular/Opercular Cortex<br>Basal Ganglia<br>Other Cerebral Cortex | Cerebellum<br>Cingulate Cortex<br>DLPC<br>Precentral<br>Cingulate Cortex<br>Other Cerebral Cortex | Cingulate Cortex<br>Inferior Frontal Gyrus<br>DLPC<br>Precentral<br>Occipital Cortex<br>Insular/Opercular Cortex<br>Basal Ganglia<br>Other Cerebral Cortex |
| <b>sub-pd015</b> | -                                                                                                                                | -                                                                     | -                                                                                                                                         | -                                                                       | -                                                                                                                                                    | -                                                                                                                              | -                                                                                                 | -                                                                                                                                                          |
| <b>sub-pd017</b> | -                                                                                                                                | -                                                                     | -                                                                                                                                         | -                                                                       | -                                                                                                                                                    | -                                                                                                                              | -                                                                                                 | -                                                                                                                                                          |
| <b>sub-pd018</b> | vmPFC/dmPFC<br>Occipital Cortex                                                                                                  | -                                                                     | vmPFC/dmPFC<br>Parietal Cortex<br>Precentral                                                                                              | -                                                                       | Inferior Frontal Gyrus<br>Precentral<br>Occipital Cortex                                                                                             | -                                                                                                                              | -                                                                                                 | Occipital Cortex                                                                                                                                           |
| <b>sub-pd020</b> | -                                                                                                                                | -                                                                     | Thalamus                                                                                                                                  | -                                                                       | -                                                                                                                                                    | -                                                                                                                              | -                                                                                                 | -                                                                                                                                                          |
| <b>sub-pd023</b> | Inferior Frontal Gyrus<br>DLPC<br>Precentral<br>Temporal Cortex<br>Thalamus<br>Insular/Opercular Cortex<br>Other Cerebral Cortex | Thalamus<br>Other Cerebral Cortex                                     | Cerebellum<br>Cingulate Cortex<br>Inferior Frontal Gyrus<br>DLPC<br>Precentral<br>Temporal Cortex<br>Thalamus<br>Insular/Opercular Cortex | Inferior Frontal Gyrus<br>Precentral<br>Other Cerebral Cortex           | Amygdala<br>Cerebellum<br>DLPC<br>Precentral<br>Hippocampus<br>Other Cerebral Cortex                                                                 | Amygdala<br>Cerebellum<br>Cingulate Cortex<br>DLPC<br>Thalamus<br>Hippocampus<br>Other Cerebral Cortex                         | Cerebellum<br>Inferior Frontal Gyrus<br>DLPC<br>Precentral<br>Temporal Cortex                     | DLPC<br>Precentral<br>Thalamus                                                                                                                             |
| <b>sub-pd024</b> | Precentral                                                                                                                       | Thalamus<br>Hippocampus                                               | Precentral                                                                                                                                | -                                                                       | -                                                                                                                                                    | -                                                                                                                              | Precentral<br>Occipital Cortex                                                                    | -                                                                                                                                                          |
| <b>sub-pd028</b> | -                                                                                                                                | -                                                                     | -                                                                                                                                         | -                                                                       | -                                                                                                                                                    | -                                                                                                                              | -                                                                                                 | -                                                                                                                                                          |

Table S8: Significant ROIs by subject and GVS session in the ON-sham ON comparison ( $FDR < 0.05$ ). **Purple** text indicates ROIs also significant in the OFF-sham OFF comparison. DLPC = dorsolateral prefrontal cortex.

| Subject   | GVS1                     | GVS2                     | GVS3                     | GVS4                                                   | GVS5                                                           | GVS6                                             | GVS7                                                                                                                                                       | GVS8                                                                                                                                                               |
|-----------|--------------------------|--------------------------|--------------------------|--------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sub-pd004 | -                        | -                        | Parietal Cortex          | Amygdala                                               | -                                                              | Hippocampus                                      | -                                                                                                                                                          | Hippocampus                                                                                                                                                        |
| sub-pd007 | Cerebellum               | Inferior Frontal Gyrus   | Amygdala                 | Hippocampus                                            | Cerebellum<br>Thalamus<br>Hippocampus<br>Other Cerebral Cortex | Hippocampus                                      | Cerebellum<br>Insular/Opercular Cortex<br>Inferior Frontal Gyrus<br>vmPFC/dmPFC<br>Hippocampus                                                             | Inferior Frontal Gyrus<br>Amygdala<br>Temporal Cortex<br>Occipital Cortex<br>Hippocampus                                                                           |
|           | Cingulate Cortex         | Cerebellum               |                          |                                                        |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Inferior Frontal Gyrus   | Inferior Frontal Gyrus   |                          |                                                        |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | DLPC                     | Temporal Cortex          |                          |                                                        |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Temporal Cortex          | Occipital Cortex         |                          |                                                        |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Occipital Cortex         | Hippocampus              |                          |                                                        |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Insular/Opercular Cortex | Other Cerebral Cortex    |                          |                                                        |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
| sub-pd009 | -                        | -                        | -                        | -                                                      | -                                                              | -                                                | -                                                                                                                                                          | -                                                                                                                                                                  |
| sub-pd010 | -                        | -                        | -                        | Thalamus                                               | Occipital Cortex                                               | -                                                | -                                                                                                                                                          | -                                                                                                                                                                  |
| sub-pd011 | vmPFC/dmPFC              | Parietal Cortex          | vmPFC/dmPFC              | Amygdala                                               | Amygdala                                                       | Occipital Cortex                                 | Cingulate Cortex<br>vmPFC/dmPFC<br>Parietal Cortex<br>Temporal Cortex                                                                                      | vmPFC/dmPFC                                                                                                                                                        |
|           | Temporal Cortex          | Thalamus                 | Cingulate Cortex         | Cerebellum                                             |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Occipital Cortex         | Occipital Cortex         | Parietal Cortex          | Occipital Cortex                                       |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Insular/Opercular Cortex | Insular/Opercular Cortex | Insular/Opercular Cortex | Insular/Opercular Cortex                               |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Insular/Opercular Cortex | Other Cerebral Cortex    | Basal Ganglia            | Basal Ganglia                                          |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
| sub-pd012 | Cingulate Cortex         | Cingulate Cortex         | Cingulate Cortex         | Amygdala                                               | Amygdala                                                       | Occipital Cortex<br>Hippocampus<br>Basal Ganglia | Amygdala<br>Cingulate Cortex<br>Inferior Frontal Gyrus<br>vmPFC/dmPFC<br>Parietal Cortex<br>Precentral<br>Occipital Cortex<br>Hippocampus<br>Basal Ganglia | Amygdala<br>Cingulate Cortex<br>Inferior Frontal Gyrus<br>DLPC<br>vmPFC/dmPFC<br>Parietal Cortex<br>Precentral<br>Occipital Cortex<br>Hippocampus<br>Basal Ganglia |
|           | Inferior Frontal Gyrus   | DLPC                     | Inferior Frontal Gyrus   | Cingulate Cortex                                       | Cingulate Cortex                                               |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Parietal Cortex          | Parietal Cortex          | Parietal Cortex          | Inferior Frontal Gyrus                                 | Inferior Frontal Gyrus                                         |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Hippocampus              | Hippocampus              | Thalamus                 | Occipital Cortex                                       | DLPC                                                           |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Other Cerebral Cortex    | Basal Ganglia            | Occipital Cortex         | Hippocampus                                            | Precentral                                                     |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Other Cerebral Cortex    | Other Cerebral Cortex    | Basal Ganglia            | Basal Ganglia                                          | Basal Ganglia                                                  |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
| sub-pd013 | -                        | -                        | -                        | -                                                      | -                                                              | -                                                | -                                                                                                                                                          | -                                                                                                                                                                  |
| sub-pd014 | Precentral               | vmPFC/dmPFC              | Precentral               | Occipital Cortex<br>Parietal Cortex<br>Temporal Cortex | Other Cerebral Cortex<br>Parietal Cortex                       | Other Cerebral Cortex<br>Parietal Cortex         | -<br>-<br>-                                                                                                                                                | Precentral<br>Occipital Cortex<br>Parietal Cortex                                                                                                                  |
|           | Parietal Cortex          | Precentral               |                          |                                                        |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
|           | Insular/Opercular Cortex | Insular/Opercular Cortex |                          |                                                        |                                                                |                                                  |                                                                                                                                                            |                                                                                                                                                                    |
| sub-pd015 | -                        | -                        | -                        | Insular/Opercular Cortex                               | -                                                              | -                                                | -                                                                                                                                                          | -                                                                                                                                                                  |
| sub-pd017 | -                        | Occipital Cortex         | DLPC                     | DLPC                                                   | -                                                              | -                                                | -                                                                                                                                                          | -                                                                                                                                                                  |
| sub-pd018 | -                        | Other Cerebral Cortex    | Insular/Opercular Cortex | Insular/Opercular Cortex                               | -                                                              | -                                                | DLPC                                                                                                                                                       | -                                                                                                                                                                  |
| sub-pd020 | -                        | -                        | -                        | -                                                      | -                                                              | -                                                | Other Cerebral Cortex<br>Cerebellum                                                                                                                        | Other Cerebral Cortex                                                                                                                                              |
| sub-pd023 | -                        | -                        | -                        | -                                                      | -                                                              | -                                                | -                                                                                                                                                          | -                                                                                                                                                                  |
| sub-pd024 | Occipital Cortex         | -                        | Occipital Cortex         | -                                                      | -                                                              | -                                                | -                                                                                                                                                          | -                                                                                                                                                                  |
|           | Occipital Cortex         | -                        | Other Cerebral Cortex    | -                                                      | -                                                              | -                                                | Occipital Cortex<br>Other Cerebral Cortex                                                                                                                  | Occipital Cortex<br>Other Cerebral Cortex                                                                                                                          |
| sub-pd028 | Amygdala                 | Insular/Opercular Cortex | Amygdala                 | Amygdala                                               | Cingulate Cortex                                               | Cingulate Cortex                                 | Inferior Frontal Gyrus                                                                                                                                     | Amygdala                                                                                                                                                           |
|           | Cingulate Cortex         | Other Cerebral Cortex    | Cingulate Cortex         | Cingulate Cortex                                       | Insular/Opercular Cortex                                       | Insular/Opercular Cortex                         | Insular/Opercular Cortex                                                                                                                                   | Cingulate Cortex                                                                                                                                                   |
|           | Insular/Opercular Cortex | Other Cerebral Cortex    | Insular/Opercular Cortex | Precentral                                             | Insular/Opercular Cortex                                       | Insular/Opercular Cortex                         | Other Cerebral Cortex                                                                                                                                      | Insular/Opercular Cortex                                                                                                                                           |
|           | Insular/Opercular Cortex | Other Cerebral Cortex    | Other Cerebral Cortex    | Insular/Opercular Cortex                               | Other Cerebral Cortex                                          | Other Cerebral Cortex                            | Other Cerebral Cortex                                                                                                                                      | Other Cerebral Cortex                                                                                                                                              |
|           | Other Cerebral Cortex    | Other Cerebral Cortex    | Other Cerebral Cortex    | Other Cerebral Cortex                                  | Other Cerebral Cortex                                          | Other Cerebral Cortex                            | Other Cerebral Cortex                                                                                                                                      | Other Cerebral Cortex                                                                                                                                              |